

SAFETY QUIZ: CHE381- Chemical Engineering Laboratory

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?

awareness

2. What two pieces of personal protective equipment must be worn at all times in the lab?

Safety glasses and shoes

3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?

Gloves, lab coats, splash aprons, face shields, ear protection

4. True or False: Fume hoods are generally used as storage areas for chemicals.

false

5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:

1000 ppm or 1000 mg/M3

6. What are the four routes of chemical entry into the body?

inhalation, ingestion, injection, and eye or skin contact

7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

false

8.  What is the role of the Chemical Hygiene Plan?

To describe the safety program and to coordinate safety procedures

9. What is the guideline for laboratory room airflow and fume hood airflow?

Lab room: ventilation not less than 20 linear feet per minute airflow

Fume hood: 80-120 linear feet per minute airflow

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

Keep all aisles, hallways, and stairs clear of all chemicals.

Keep all work areas and workbenches clear of clutter and obstructions.

All working surfaces and floors should be cleaned regularly.

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11. What is the proper method of transporting gas cylinders in the laboratory?

Use 1-cylinder, 4-wheel carts, with cylinder cap on

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

false

13. What is an accepted method of vapor detection in the laboratory?

The use of mechanical vapor detectors

14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.

Wash promptly if skin contact is made with any chemical, regardless of corrosivity.

Wear appropriate eye protection at all times.

Avoid inhalation of chemicals, do not "sniff" test chemicals.

15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.

Use bottle carrier that is shock resistant and prevents breakage such as Nalgene Safety Bottle Carrier, made of LDPE, (cat. # 03-439-29) and Safety Bottle Carrier (cat. # 03-439).

16. List the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.

Evacuate lab, Dial "O", describe emergency and location to operator, evacuate building if necessary. Operator activates emergency response team.

17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

Flammable: chemicals with a flash point below 100 degrees F.

Combustible: chemicals with flash points between 100 and 200 degrees F.

18. List the Emergency Procedure for an individual chemical exposure in the laboratory.

Get to safety shower/eyewash ASAP, flush affected body with water for at least 15 minutes. If alone, activate emergency button, otherwise co-worker dials "O" and describes emergency and location. Emergency response team will be activated and respond.

19. What is the definition of a "hazardous chemical"?

hazardous chemical: a chemical with statistically significant evidence that an acute or chronic health effect may occur in exposed employees.

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

eyewashes: 3 gal/ min, checked routinely

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safety showers: 30 gal/ min, checked routinely/once a month